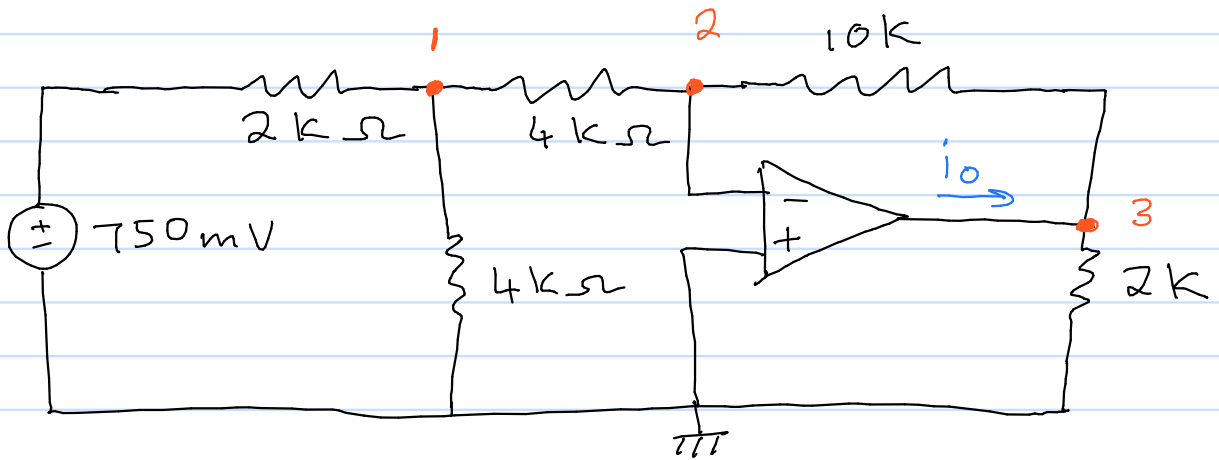


5.19 Determine i_o



At node 1: KCL

$$\frac{0,75 - V_1}{2000} = \frac{V_1}{4000} + \frac{V_1}{4000}$$

$$\frac{0,75 - V_1}{2000} = \frac{V_1}{2000}$$

$$0,75 = 2V_1$$

$$V_1 = 0,375$$

At node 2: KCL

$$\frac{0,375}{4000} = \frac{-V_o}{10000}$$

$$V_o = \frac{-0,375 \times 10000}{4000}$$

$$V_o = -0,9375$$

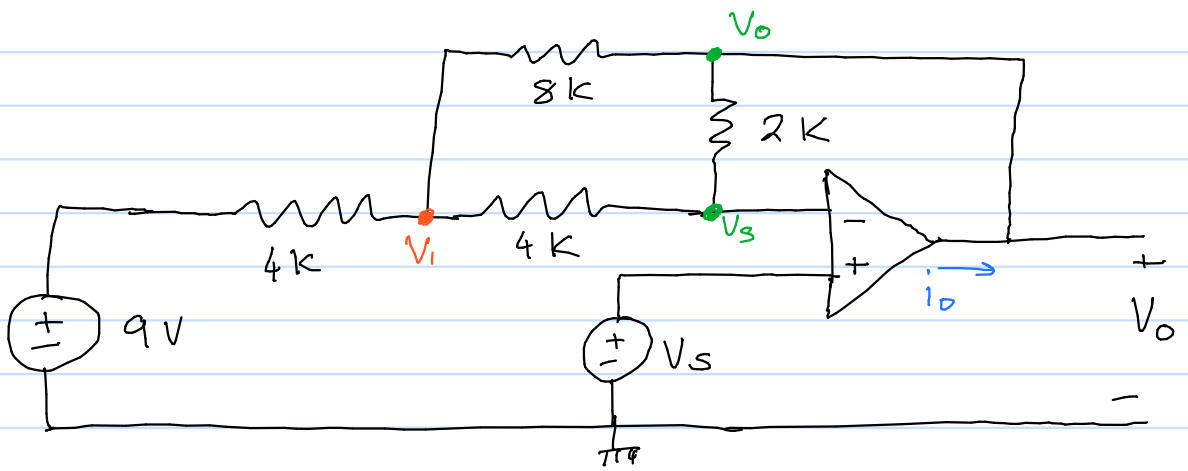
At node 3: KCL

$$\frac{0,375 - (-0,9375)}{14000} + i_o = \frac{-0,9375}{2000}$$

$$i_o = -468,75 \mu A - 93,75 \mu A$$

$$i_o = -562,5 \mu A$$

5.20 Calculate V_o if $V_s = 2\text{ V}$



KCL at V_1 :

$$\frac{9 - V_1}{4000} + \frac{V_s - V_1}{4000} + \frac{V_o - V_1}{8000} = 0 \quad (1)$$

KCL at V_o :

$$\frac{V_1 - V_o}{8000} + \frac{V_s - V_o}{2000} + i_o = 0 \quad (2)$$

KCL at V_s :

$$\frac{V_1 - V_s}{4000} + \frac{V_o - V_s}{2000} = 0 \quad (3)$$

* Equation 1 & 3 are sufficient to find V_o .

$$18 - 2V_1 + 2V_s - 2V_1 + V_o - V_1 = 0 \quad (1)$$

$$18 - 2V_1 + 2(2) - 2V_1 + V_o - V_1 = 0$$

$$V_o - 5V_1 = -22$$

$$V_1 - V_s + 2V_o - 2V_s = 0$$

$$V_1 - 2 + 2V_o - 2(2) = 0$$

$$2V_o + V_1 = 6$$

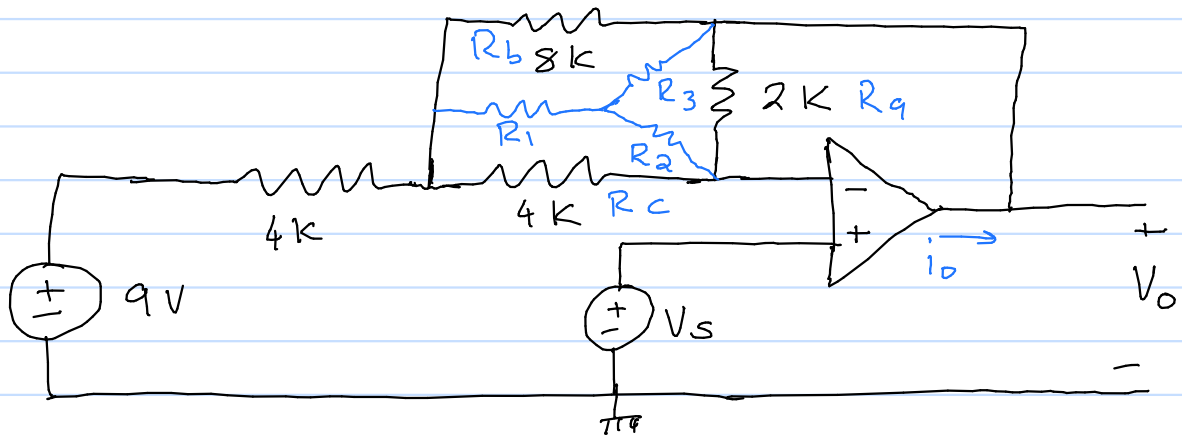
Set up and solve matrix equation

$$\begin{bmatrix} 1 & -5 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} V_o \\ V_1 \end{bmatrix} = \begin{bmatrix} -22 \\ 6 \end{bmatrix}$$

$$V_o = 0.727273 = 727.273\text{ mV}$$

$$V_1 = 4.545\text{ V}$$

5.20 Calculate V_o if $V_s = 2\text{ V}$



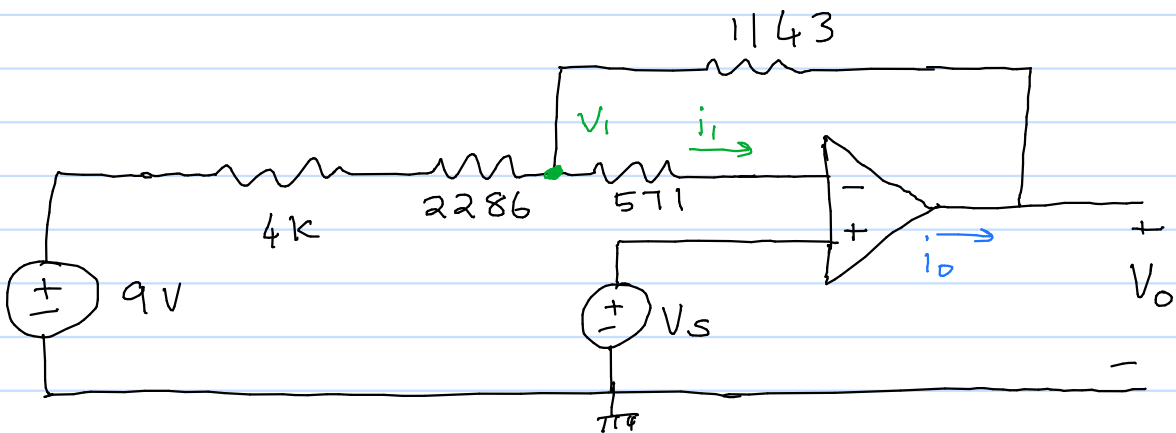
Δ -Y transform

$$R_a + R_b + R_c = 2000 + 8000 + 4000 = 14000$$

$$R_1 = \frac{R_b R_c}{R_a + R_b + R_c} = \frac{8000 \times 4000}{14000} = 2286 \Omega$$

$$R_2 = \frac{R_c R_a}{R_a + R_b + R_c} = \frac{4000 \times 2000}{14000} = 571 \Omega$$

$$R_3 = \frac{R_a R_b}{R_a + R_b + R_c} = \frac{2000 \times 8000}{14000} = 1143 \Omega$$



Since $i_i = 0$, $V_i = V^- = V_s$

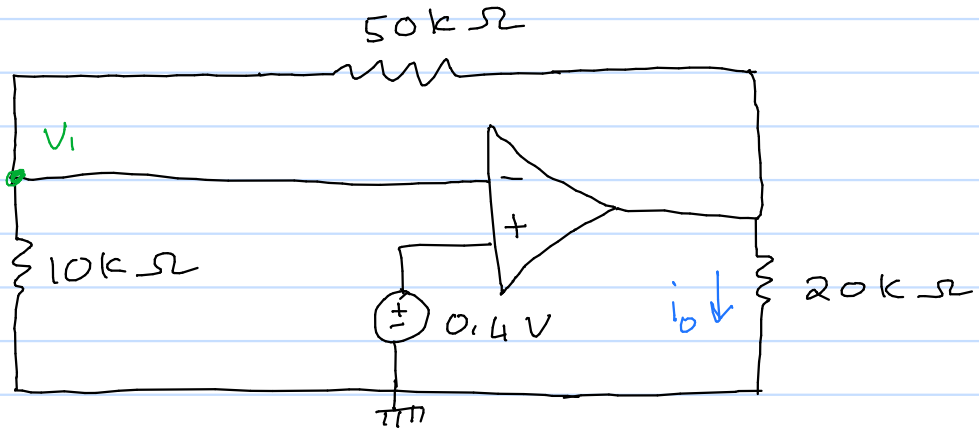
$$\frac{9 - V_s}{6286} = \frac{V_s - V_o}{1143}$$

$$\frac{9 - 2}{6286} = \frac{2 - V_o}{1143}$$

$$V_o = \left(\frac{7}{6286} - \frac{2}{1143} \right) (-1143)$$

$$V_o = 0.727 \text{ V} \\ = 727.171 \text{ mV}$$

5,28 Find i_o



KCL at node V_1 :

$$\frac{-V_1}{10000} = \frac{V_1 - V_o}{50000}$$

$$\frac{-0.4}{10000} = \frac{0.4 - V_o}{50000}$$

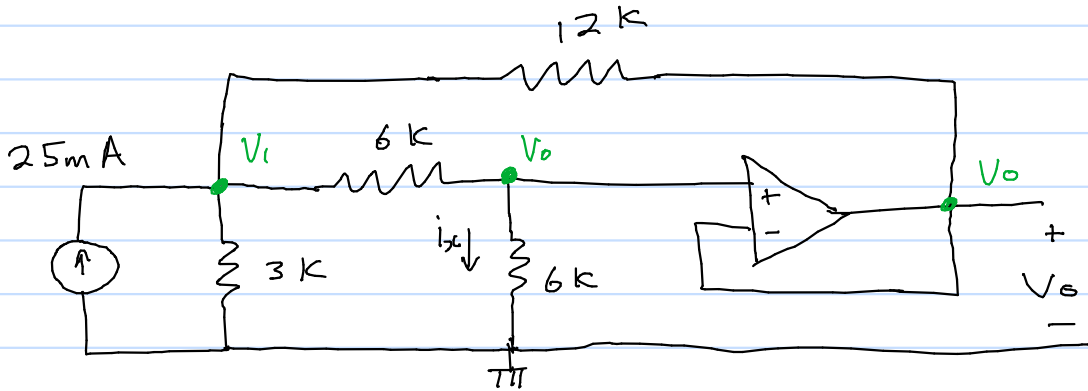
$$V_o = \left(\frac{-0.4}{10000} - \frac{0.4}{50000} \right) (-50000)$$

$$V_o = 2.4 \text{ V}$$

$$i_o = \frac{V_o}{20000}$$

$$i_o = 120 \mu\text{A}$$

5.31 find i_x



KCL at V_1

$$0.025 = \frac{V_1}{3000} + \frac{V_1 - V_0}{6000} + \frac{V_1 - V_0}{12000}$$

$$300 = 4V_1 + 2V_1 - 2V_0 + V_1 - V_0$$

$$300 = -3V_0 + 7V_1$$

KCL at V_0

$$\frac{V_1 - V_0}{6000} + \frac{0 - V_0}{6000} = 0$$

$$V_1 - V_0 - V_0 = 0$$

$$-2V_0 + V_1 = 0$$

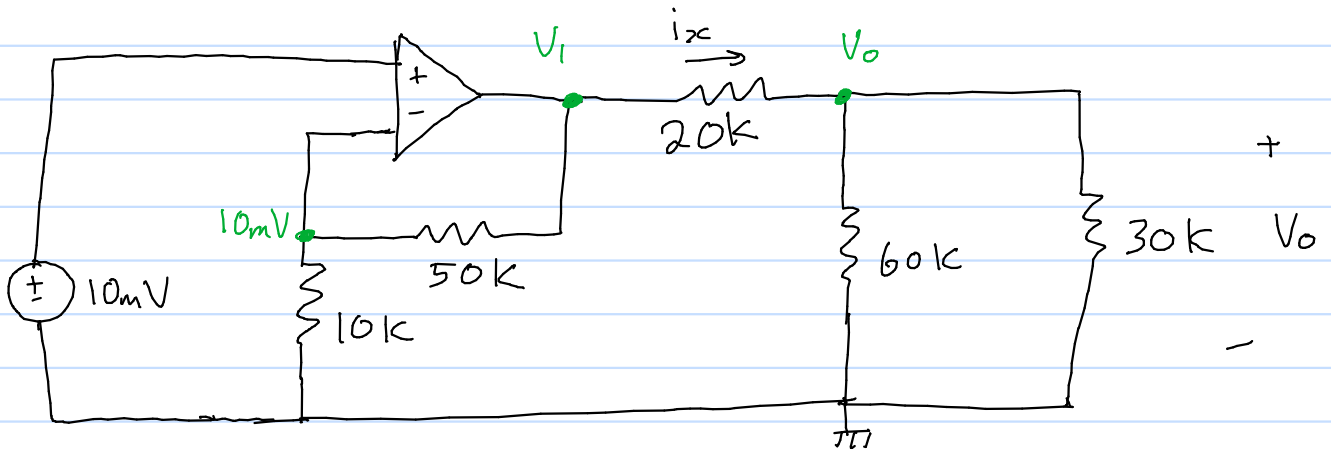
Set up and solve matrix equation

$$\begin{bmatrix} -3 & 7 \\ -2 & 1 \end{bmatrix} \begin{bmatrix} V_0 \\ V_1 \end{bmatrix} = \begin{bmatrix} 300 \\ 0 \end{bmatrix}$$

$$V_0 = 27.273 \text{ V}$$

$$V_1 = 54.545 \text{ V}$$

5.32 Calculate i_x , V_o and the power dissipated in the $60\text{k}\Omega$ resistor



KCL at 10mV node:

$$\frac{V_1 - 0.01}{5000} = \frac{0.01}{10000}$$

$$V_1 = \left(\frac{0.01}{10000} + \frac{0.01}{50000} \right) 50000$$

$$V_1 = 0.06$$

Find V_o through voltage division

$$V_o = \frac{V_1 \times 60000 \parallel 30000}{60000 \parallel 30000 + 20000}$$

$$V_o = 0.03 = 30\text{ mV}$$

$$i_x = \frac{V_1 - V_o}{20000}$$

$$= \frac{0.06 - 0.03}{20000}$$

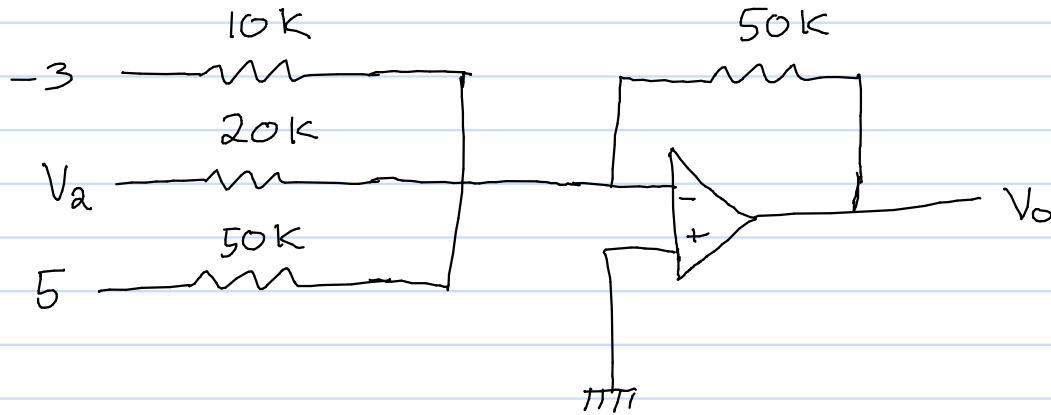
$$= 1.5\text{ }\mu\text{A}$$

$$P_{60\text{k}} = \frac{V_o^2}{60000}$$

$$= \frac{0.03^2}{60000}$$

$$= 15\text{ nW}$$

5,39 find V_2 such that $V_o = -16.5$ V



Summing amplifier

$$V_o = -\left(\frac{50000}{10000} \times (-3) + \frac{50000}{20000} V_2 + \frac{50000}{50000} \times 5 \right)$$

$$-16.5 = 15 - \frac{5}{2} V_2 - 5$$

$$V_2 = (-16.5 - 15 + 5) \times \frac{-2}{5}$$

$$V_2 = 10.6 \text{ V}$$

5.45 Design an op-amp circuit to perform the following operation:

$$V_o = 3.5 V_1 - 2.5 V_2$$

* Difference amplifier

$$V_o = \frac{R_2 (1 + R_1/R_2)}{R_1 (1 + R_3/R_4)} V_2 - \frac{R_2}{R_1} V_1$$

$$\frac{R_2}{R_1} = 2.5$$

$$\frac{2.5 (1 + 2.5^{-1})}{1 + R_3/R_4} = 3.5$$

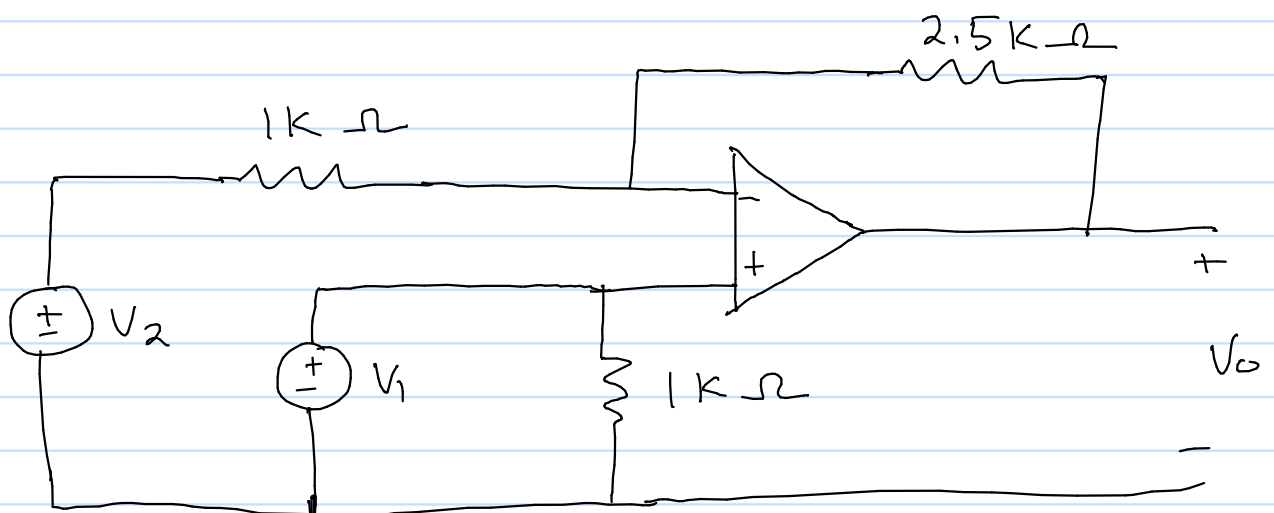
$$2.5 (1.4) = 3.5 + \frac{R_3}{R_4} 3.5$$

$$3.5 - 3.5 = \frac{R_3}{R_4} 3.5$$

$$\frac{R_3}{R_4} = 0$$

$$R_1 = 1 \text{ k}\Omega \quad R_2 = 2.5 \text{ k}\Omega$$

$$R_3 = 0 \Omega \quad R_4 = 1 \text{ k}\Omega$$



5.45 Design an op-amp circuit to perform the following operation:

$$V_o = 3.5 V_1 - 2.5 V_2$$

* Summing amplifier

$$V_o = - \left(\frac{R_f}{R_1} V_1 + \frac{R_f}{R_2} V_2 \right)$$

* Inverting amplifier

$$V_o = - \frac{R_f}{R_i} V_i$$

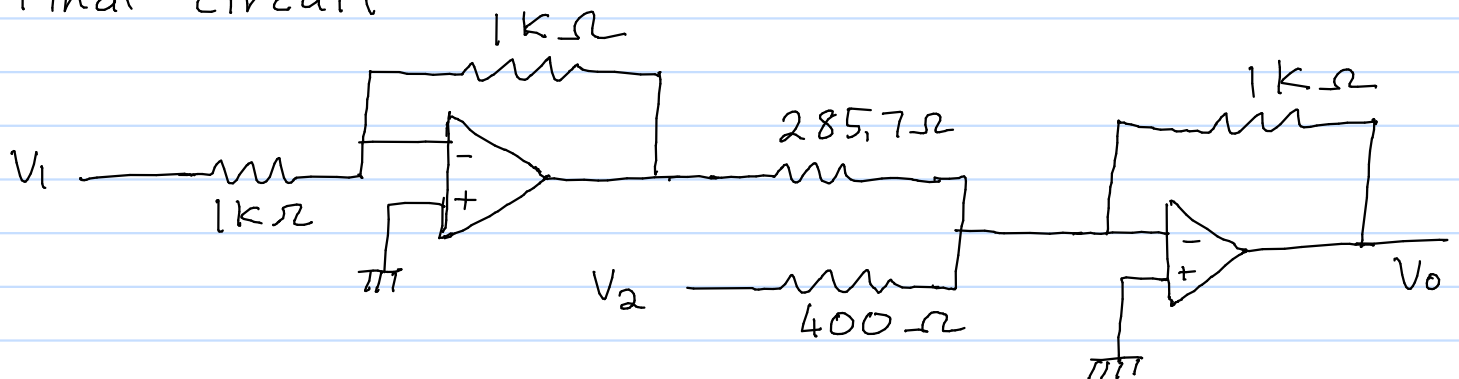
$$V_o = 3.5 V_1 - 2.5 V_2$$

$$= -(-3.5 V_1 + 2.5 V_2)$$

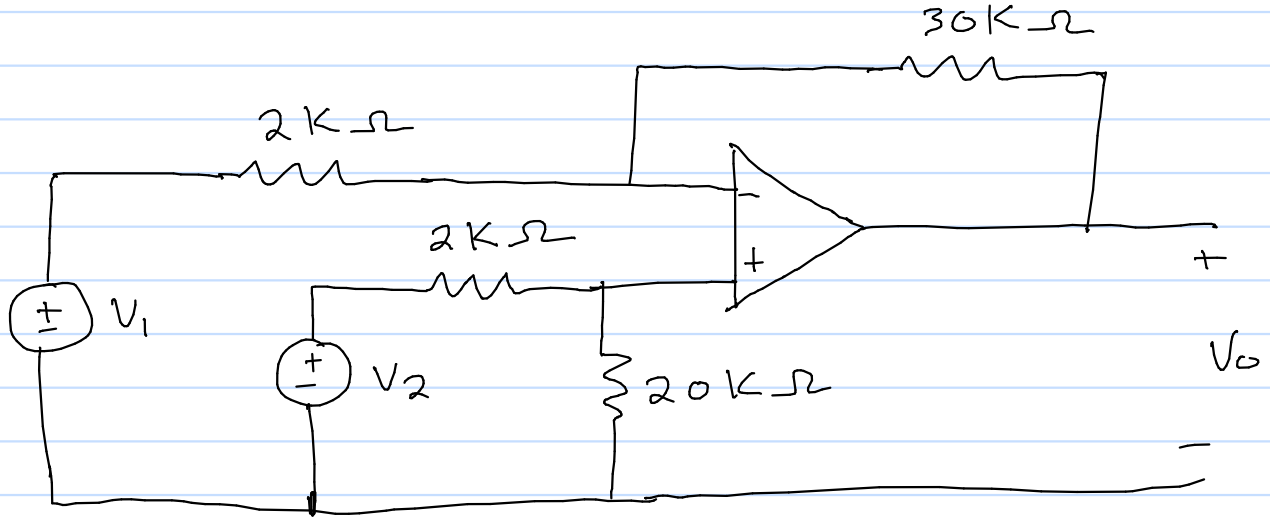
$$= - \left[\frac{3500}{1000} \cdot \left(\frac{-1}{1} \right) V_1 + \frac{2500}{1000} V_2 \right]$$

$$= - \left[\frac{1000}{285.714} \left(\frac{-1000}{1000} \right) V_1 + \frac{1000}{400} V_2 \right]$$

Final circuit



S147 Find V_o , given $V_1 = 1\text{ V}$ & $V_2 = 2\text{ V}$

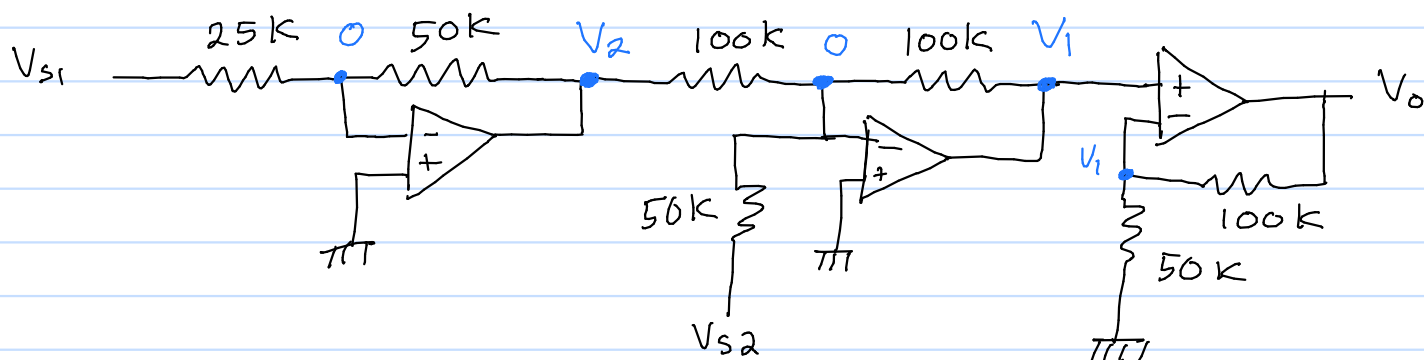


$$V_o = \frac{R_2 (1 + R_1/R_2)}{R_1 (1 + R_3/R_4)} V_2 - \frac{R_2}{R_1} V_1$$

$$V_o = \frac{15 (1 + 1/15)}{1 + 1/10} V_2 - 15 V_1$$

$$V_o = 14.545 V_2 - 15 V_1$$

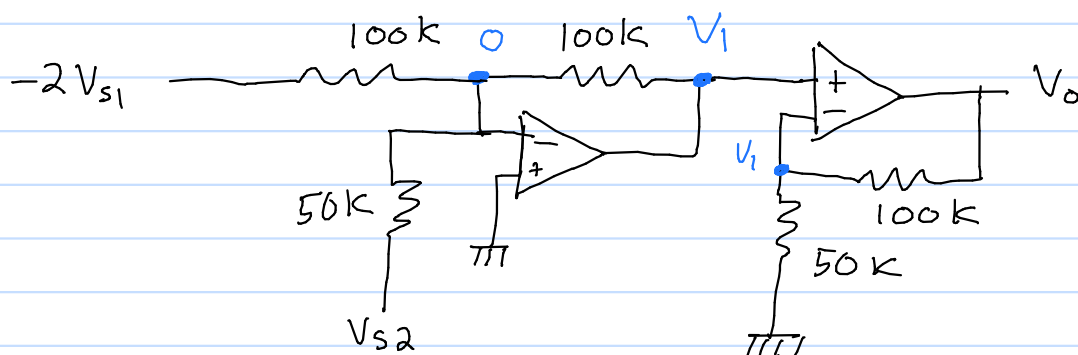
5.57 Find V_o



Find V_2 :

$$\frac{V_{s1}}{25000} = \frac{-V_2}{50000}$$

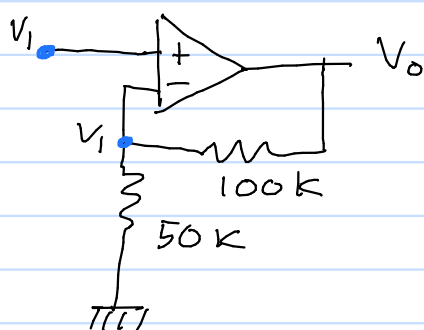
$$V_2 = -2V_{s1}$$



Find V_1

$$\frac{-2V_{s1}}{100000} + \frac{V_{s2}}{50000} = \frac{-V_1}{100000}$$

$$V_1 = 2V_{s1} - 2V_{s2}$$



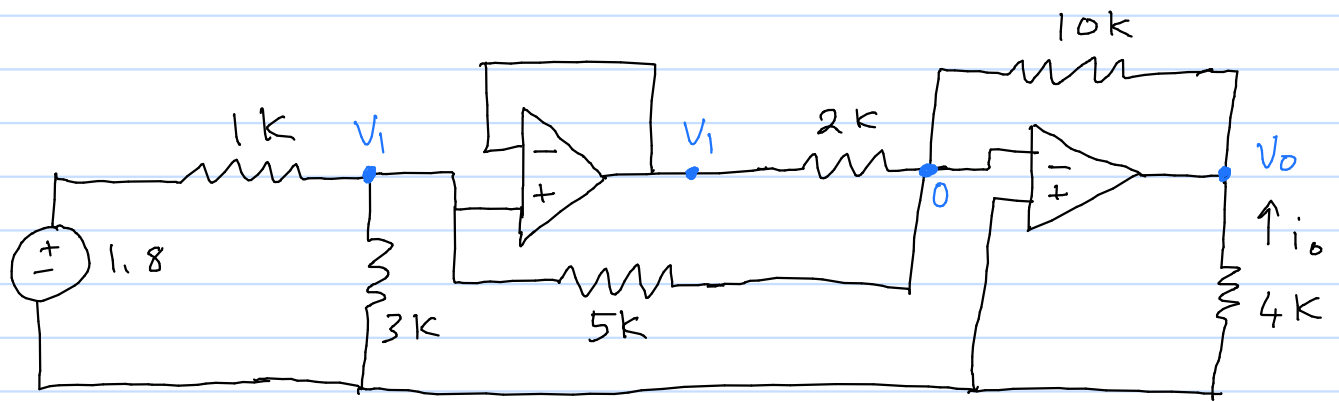
$$V_1 = \frac{V_o \times 50000}{50000 + 100000}$$

Voltage division

$$V_o = 3V_1$$

$$V_o = 6V_{s1} - 6V_{s2}$$

5.58 find i_o



KCL at V_1 :

$$\frac{1.8 - V_1}{1000} = \frac{V_1}{3000} + \frac{V_1}{5000}$$

$$\frac{1.8}{1000} = V_1 \left(\frac{1}{1000} + \frac{1}{3000} + \frac{1}{5000} \right)$$

$$0.0018 = V_1 \left(\frac{23}{15000} \right)$$

$$V_1 = \frac{27}{23}$$

KCL at 0:

$$\frac{V_1}{2000} + \frac{V_1}{5000} = \frac{-V_o}{10000}$$

$$5V_1 + 2V_1 = -V_o$$

$$V_o = -7V_1$$

$$V_o = -7 \times \frac{27}{23}$$

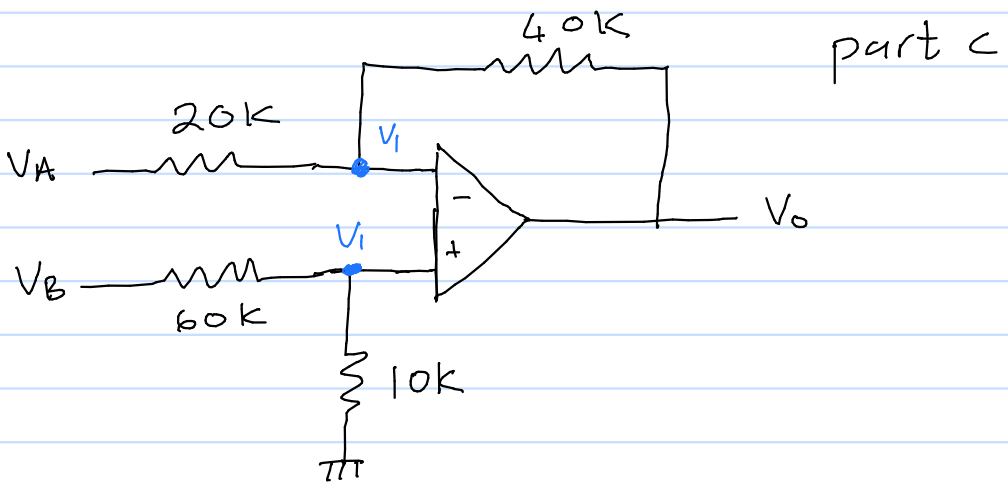
$$V_o = \frac{-189}{23}$$

$$i_o = \frac{-V_o}{4000}$$

$$= 2.054 \text{ mA}$$

5.70 find V_o

Seperate parts A, B, C and solve each 1:

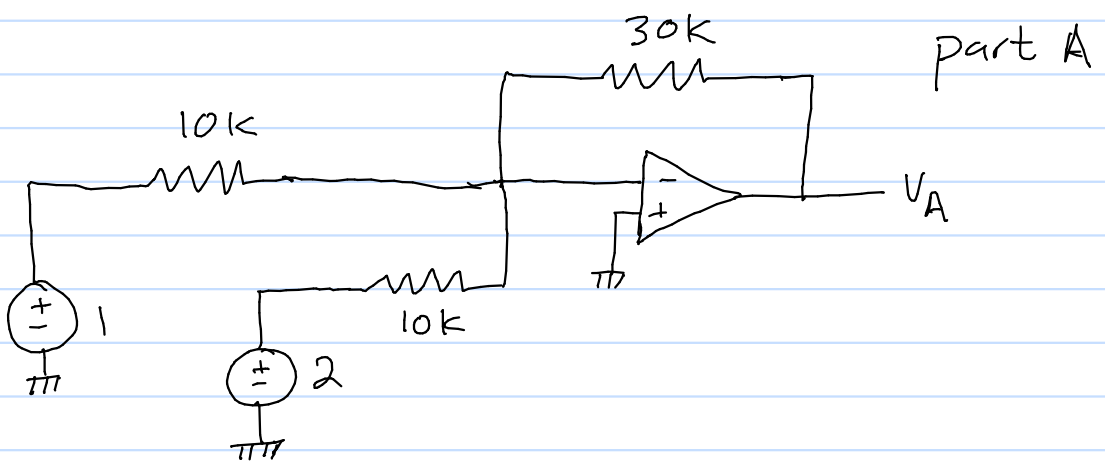


$$V_1 = \frac{V_B \times 10\,000}{10\,000 + 60\,000} = \frac{V_B}{7}$$

$$\frac{V_A - V_B/7}{20\,000} = \frac{V_B/7 - V_o}{40\,000}$$

$$2V_A - \frac{2V_B}{7} = \frac{V_B}{7} - V_o$$

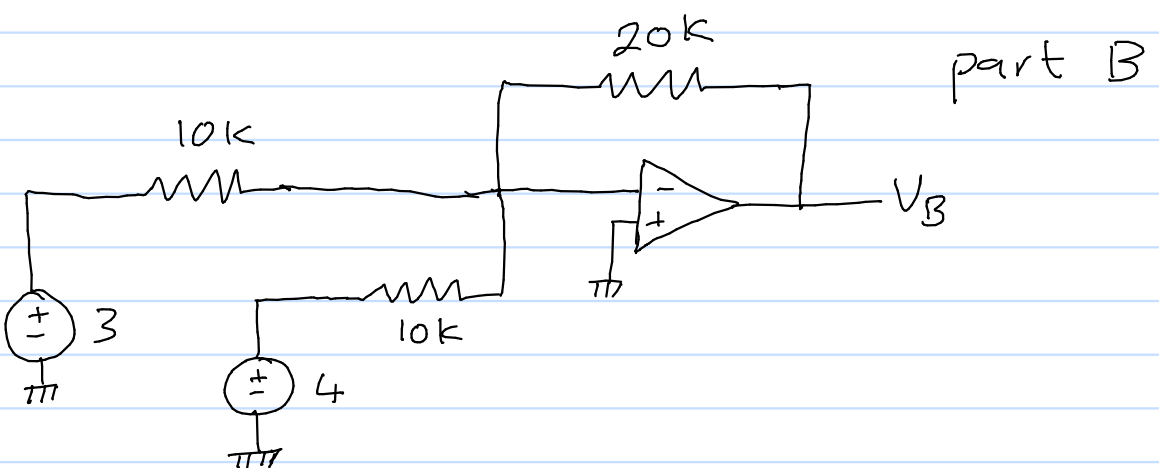
$$V_o = \frac{3V_B}{7} - 2V_A$$



* Summing amplifier

$$V_A = - \left[\frac{30k}{10k}(1) + \frac{30k}{10k}(2) \right]$$

$$= -9\text{ V}$$



* Summing amplifier

$$V_B = - \left[\frac{20k}{10k}(3) + \frac{20k}{10k}(4) \right]$$

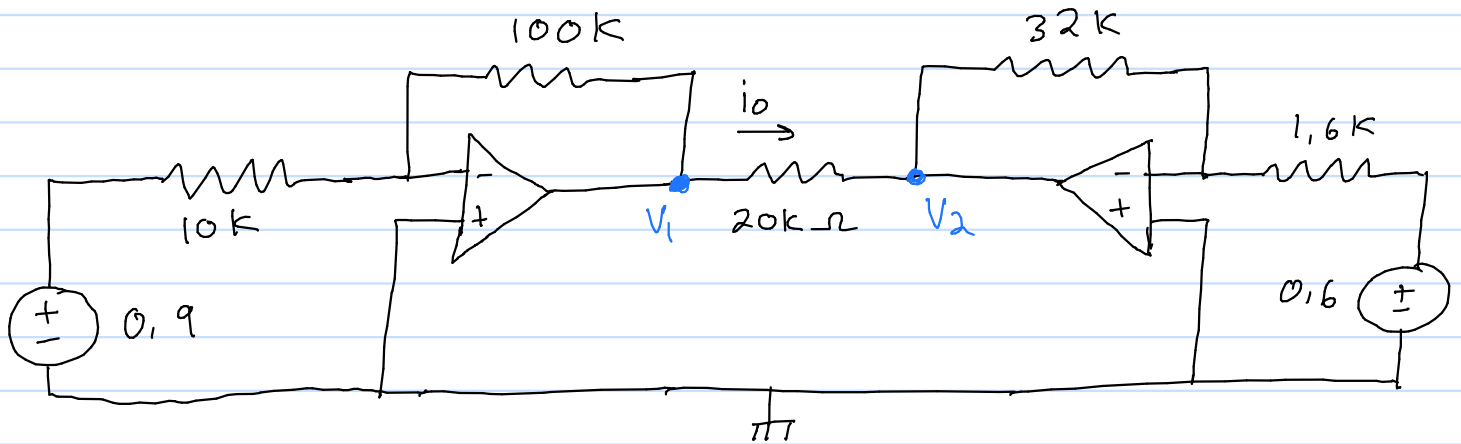
$$= -14\text{ V}$$

$$V_o = \frac{3V_B}{7} - 2V_A$$

$$V_o = \frac{-3 \times 14}{7} - 2 \times (-9)$$

$$V_o = 12\text{ V}$$

5.74 Find i_o



* Inverting amplifier

$$V_1 = -0.9 \times \frac{100k}{10k}$$

$$= -9V$$

$$V_2 = -0.6 \times \frac{32k}{1.6k}$$

$$= -12V$$

$$i_o = \frac{V_1 - V_2}{20k}$$

$$= \frac{-9 - (-12)}{20k}$$

$$= 150\mu A$$